

Amirhossein Arefzadeh

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Education

B.S. in Computer Engineering

Sep 2022 – Present

University of Tehran

- GPA: 19.01/20.0 (3.91/4) (Faculty Average: 14.96/20)
- 1st Rank ◇ Spring 2024 Semester ◇ 19.83/20.0
- Ranked 6th out of 90 students

Highschool Diploma in Mathematics

Sep 2019 – Jun 2022

Ario Mosallanezhad, Mashhad

- GPA: 19.6/20.0

Related Courses

- Artificial Intelligence: 19.5/20
- Machine Learning: 19.3/20
- Computer Networks: 20/20
- Engineering Probability and Statistics: 17.7/20
- Basic Computer Programming: 20/20
- Discrete Mathematics: 20/20
- Foundations of Data Science: 19.8/20
- Deep generative models (MSc): 19.4/20
- Algorithm Design: 19.25/20
- Database Design: 18.7/20
- Advanced Programming: 19.6/20
- Data Structures and Algorithms: 20/20

Research Interests

- Trustworthy ML
- LLMs
- Fairness and Robustness
- Computational Neuroscience

Research Experience

◦ Research Assistant, Machine Unlearning

July 2025 – Nov 2025

RIML (Robust and Interpretable Machine Learning Lab)

Under the supervision of [Dr. M. H. Rohban](#) and [Dr. A. M. Sadeghzadeh](#)

Contributed to cutting-edge research on robust and interpretable generative models by developing and implementing machine unlearning methods to enhance model safety and control.

◦ Investigating NLP and Brain-Inspired Models

Jan 2025 – Apr 2025

Under the supervision of [Dr. Y. Yaghoobzadeh](#)

Conducted literature review and exploratory experiments on how insights from human brain activity can improve NLP model interpretability. [[NeurIPS 2019 Background Paper](#)].

Teaching Experience

University of Tehran

Head Teaching Assistant ARTIFICIAL INTELLIGENCE Prof. H. Fadaei

Feb 2026 – Present

Teaching Assistant ARTIFICIAL INTELLIGENCE Prof. Y. Yaghoobzadeh, H. Fadaei

Feb 2025 – Feb 2026

Supervising Teaching Assistant ADVANCED PROGRAMMING Prof. R. Khosravi

Feb 2025 – Feb 2026

Supervising Teaching Assistant DISCRETE MATHEMATICS Prof. S. Mohammadi

Aug 2025 – Feb 2026

Teaching Assistant DATA STRUCTURES AND ALGORITHMS Prof. H. Faili

Aug 2024 – Feb 2026

Teaching Assistant ADVANCED PROGRAMMING Prof. R. Khosravi

Aug 2024 – Feb 2025

Teaching Assistant DISCRETE MATHEMATICS Prof. S. Mohammadi

Feb 2024 – Feb 2025

Lab Supervisor COMPUTER SYSTEMS & PROGRAMMING Profs. M. R. Hashemi, H. Moradi

Aug 2024 – Feb 2025

Teaching Assistant COMPUTER SYSTEMS & PROGRAMMING Prof. A. Kamandi

Aug 2023 – Feb 2024

Industrial Experience

- **AI Software Engineer,**
Divar - Tehran, Iran

Dec 2025 – Present

Building and maintaining production ML systems for content moderation and trust & safety at Iran's largest classifieds platform, serving 35M+ active users.

- **Review Team**

Apr 2026 – Present

Ensured the reliability of a moderation pipeline of 20+ bots that validate submitted posts, maintaining smooth end-to-end operation of the system.

Developed and improved a **category-validator model** to prevent users from listing posts in the wrong category across 200+ categories.

Worked on a **duplicate-detection bot** to prevent identical posts from being submitted more than once.

Worked on a **spam-detection bot** to identify and penalize spammers, protecting the platform from abuse.

Collaborated with a teammate to investigate an abnormally high refund rate relative to industry benchmarks, researching and proposing actionable solutions to reduce it.

- **Evaluation & Fine-tuning Team**

Dec 2025 – Apr 2026

Fine-tuned task-specific models to replace costly external API calls, eliminating network-reliability issues (critical under Iran's connectivity constraints) while cutting cost and latency and improving task-specific performance through faster, dedicated inference.

Supported other teams in evaluating their AI models, using them efficiently, and monitoring and optimizing their operational costs.

Fine-tuned, evaluated, and maintained a **field extraction model** that automatically extracts listing attributes mentioned in user-submitted posts, removing redundant form-filling steps and improving user experience.

Fine-tuned, maintained, and evaluated a **restricted-content detection model** to flag prohibited posts (e.g., illegal or underage animal sales, service requests, and medicine sales).

Fine-tuned, maintained, and evaluated a **political-content detection model** to prevent political posts, a safety-critical task given the country's situation at the time.

- **Software Engineering Intern,**
Divar - Tehran, Iran

Oct 2025 – Dec 2025

Completed an intensive training program covering backend development with Go (Golang), relational databases (PostgreSQL), and containerization and orchestration with Docker and Kubernetes, along with LLM fundamentals and building LLM-based solutions; applied these skills by contributing to real, team-owned production projects.

Projects

- **Experimental Analysis of Concept Forgetting in Generative Models** [GitHub Repository](#) 

Implemented the "Selective Amnesia" (NeurIPS 2023) paper and extended it with novel experiments on sequential forgetting, adversarial concept recovery, and alternative gradient projection methods.

- **Data Science Course Projects** [GitHub Repository](#) 

A collection of projects covering fundamental and advanced data science topics, including data scraping, data visualization, big data engineering (Kafka, Spark), CI/CD automation, machine learning (regression, classification), and deep learning (MLP, CNN, RNN).

- **Image Reconstruction with Genetic Algorithms** [GitHub Repository](#) 

Developed a Python-based genetic algorithm from scratch to recreate target images using triangles.

- **NLP News Articles Clustering** [GitHub Repository](#) 

Applied clustering techniques on news articles through text preprocessing, feature extraction, and dimen-

sionality reduction.

- **Connect4 Game Bot** [GitHub Repository](#) 

Created an intelligent Connect 4 game bot using Minimax with alpha-beta pruning. Includes an interactive interface with adjustable difficulty.

Achievements

- **Sanjesh Organization:**

Ranked 813th in Iran's National University Entrance Exam (Top 0.5%)

- **University of Tehran ACM Student Chapter:**

Completed UT ACM Summer of Code 2024 Deep Learning Course

Technical Skills

- **Languages:** Python, Go (Golang), C, C++, Java, SQL
- **ML & Data:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Apache Spark, Apache Kafka, Airflow, Trino
- **Databases & Storage:** PostgreSQL, MySQL, MongoDB, S3
- **Backend & Infrastructure:** gRPC, Docker, Kubernetes, Helm, Argo, Grafana
- **LLM Tooling:** Langfuse
- **Tools & Visualization:** Git, GitHub, Linux, Bash Script, Makefile, Metabase, Power BI, Tableau

Languages

- **Persian:** Native
- **English:** Listening: B2, Reading: C1, Writing: B1, Speaking: B2